Core Electronics

Component	Specification	Avail
Microcontroller	☑ **ESP32** (recommended for Wi-Fi, BLE, and enough GPIOs) or Arduino Mega	YES
Motor Driver	☑ **L298N Dual H-Bridge Module** (supports 2 motors)	YES
Power Switch	Toggle/rocker switch (12V/5A rated)	
Voltage Regulator	☑ **LM2596** buck converter (for regulating 12V to 5V or 3.3V)	YES
Relay Module	☑ 1-channel or 2-channel 5V relay (for switching pump)	

Motion & Chassis Components

Component	Specification	Avail
DC Geared Motors	☑ BO Motors (100/150 RPM) or 12V DC gear motors	YES
Wheels	Matched with motors, 65mm or 100mm	YES
Chassis	Acrylic, aluminum, or custom wood base	YES
Caster Wheel	1x (for support)	YES

Sensors & Navigation

Sensor	Purpose	Qty	Avail
Ultrasonic Sensors (HC-SR04)	Obstacle detection (front/side)	3	YES
IR Sensors (TCRT5000)	Edge detection / Short-range obstacle	2+	
DHT11 or DHT22	Temp + Humidity monitoring	1	YES
PIR Motion Sensor (HC-SR501)	Human detection (for safety interlock)	1	
Line Follower Sensor Array	Line tracking (preset paths)	1 module (3–5 sensors)	
Water Level Sensor	Check disinfectant level	1	

(Optional) **MQ-3/MQ-135**	Gas/alcohol/mist concentration sensor	1	

Sanitizing Mechanism

Component	Specification	Avail
Mini Water Pump	5V/6V submersible pump or diaphragm pump	
Spray Nozzle	Fog/mist spray nozzle (paired with pump)	
Disinfectant Tank	500ml – 1L plastic/glass container	
Tubing/Piping	PVC/silicone pipe (for mist direction)	
Relay Module	To control spray pump using ESP32/Arduino	

Power System

Component	Specification	Avail
Battery	☑ **12V 2200mAh Li-ion** (or 7.4V 2S if using step-up for motors)	YES
Charging Module	**TP4056** (if using Li-ion cells)	YES
Voltage Converter	☑ LM2596 (for 5V/3.3V needs from 12V)	
Power Distribution	Screw terminals or custom PCB/breadboard	
Optional Backup	9V battery for logic (not preferred, use buck converter instead)	

Assembly & Support

Component	Usage	Avail
Jumper Wires	For connections	
Breadboard or PCB	For sensor/motor/pump interfacing	YES
Nuts/Bolts	For mounting components	
Standoffs	For board separation	
Buzzer + LED	Alerts and feedback	YES
Enclosure/Shielding	For pump/sensor protection	

Optional Modules & Interface

Component	Purpose	Avail
SD Card Module	For data logging (DHT + PIR + line data)	YES
OLED Display (0.96")	Show system status (battery, spray ON/OFF)	
RTC Module (DS3231)	For time-based sanitization scheduling	